



ILLUSTRATED GAME HANDBOOK

Play with VirtualGlove

Programs 1-14, nine cartridge programs, and a whole library to rediscover.

2026 EDITION / IAIN BENNETT / MIT LICENSE

The **VirtualGlove Controller (Arduino UNO Q)** watches your hand and sends the recognized controls to RetroPie.

This guide provides game-specific play cards and explains how to use the original Programs 1-14 and cartridge Programs A-I. It shows you which gestures to make, what controls they produce, and how to try them with other games in your library.

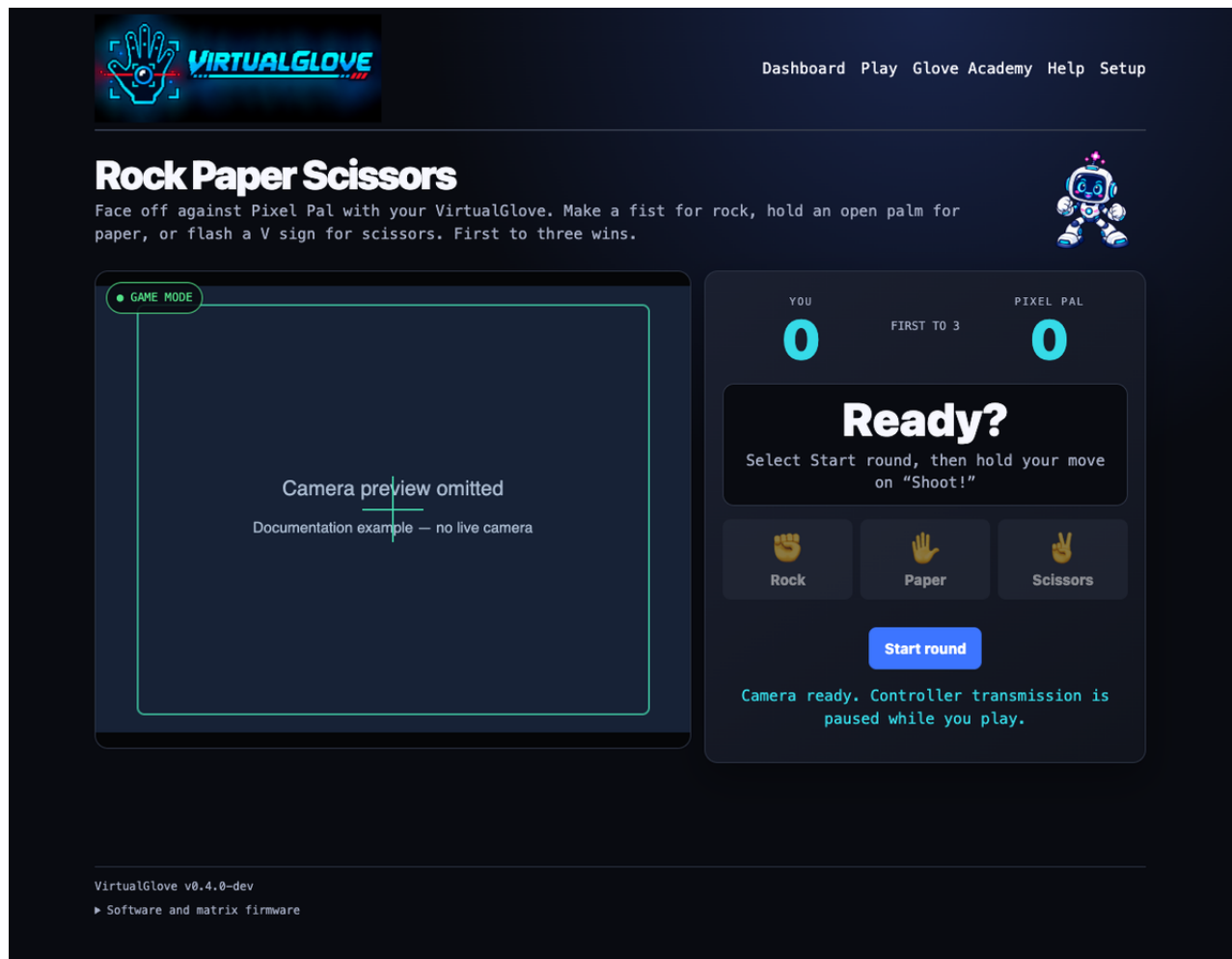
Find your game below, check its profile, and try the first-round exercise. If the system is not installed yet, start with the [Installation Guide](#).

Play Rock Paper Scissors locally

Open **Play** at <http://UNO-Q-NAME.local:8088/play> for a first-to-three match against Pixel Pal. This local game uses the Controller camera and pauses cabinet input while the page is open, so RetroPie does not need to be connected.

Your move	Make this pose	See it
Rock	Close all five fingers into a comfortable fist.	
Paper	Face an open, relaxed palm toward the camera.	
Scissors	Extend the index and middle fingers in a V sign.	

Select **Start round**, follow the countdown, and hold the pose after **Shoot!** until its button is highlighted. The first player to win three rounds takes the match. Mouse and touch buttons remain available if the camera is unavailable.



Local Rock Paper Scissors, with the camera preview omitted in this documentation example.

Get ready to play

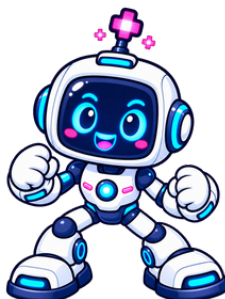
Choose your player in **Active player** so practice and sensitivity changes belong to you. Select a profile on Dashboard and wait for the camera view. Starting straight after a reboot can take longer.

1. Stand where the camera can see your whole hand, with room to move on every side.
2. Face a relaxed open palm toward the camera. On first use, or after moving the camera or changing your playing position, select **Center hand** and hold still until it finishes. Otherwise use your saved resting position.
3. Select **Start controller** when ready. If the tracker is reconnecting, Start remains pending until it can be delivered; **Stop controller** cancels that request.
4. Launch a registered game and allow its short startup pause to finish. Check the selected profile on Dashboard against the play card below; the card also shows its matrix display.
5. Try one gesture at a time. Return to your resting position between attempts. In standard movement profiles this stops directional input; Nestopia (VirtualGlove) follows your hand's position continuously.

Start controller stays armed across Controller restarts, but sends controls only while a registered game or a manually selected Dashboard profile is active. Exiting the game or losing its session releases controls. **Stop controller** keeps delivery stopped until you explicitly start it again. Use it before repositioning yourself or the camera.

Read the camera view

The overlay labels the detected hand **Right** or **Left** and shows the tracker's confidence. Dashboard's D-pad, button, and axis readings show the controls your hand produces. Keep your whole hand visible and use small, comfortable movements. If your resting hand causes unwanted movement, recalibrate in that position.



Pixel Pal takes a ready-to-play stance

Pixel Pal's Extra-Digit Hunt: Some glove illustrations have five fingers plus a thumb. Count every six-digit hand once per appearance, including repeated artwork. Pixel Pal reveals the answer at the back of this guide.

Practise with Glove Academy

Open **Glove Academy** from the main navigation to practise with Pixel Pal. Cabinet controller output pauses while you learn. The sixteen lessons cover showing your hand, finding your resting position, four movement directions, index and thumb curls, Start and Select poses, forward and backward movement, wrist rolls, a closed hand, and Menu guard.

Complete all sixteen lessons to earn **Glove Master**. The award replaces the lesson card. Progress is saved for the selected player across browsers and restarts; **Start again** clears that player's saved lesson progress and returns to the lessons. It does not clear their hand settings.

Glove Academy practises recognition, not each game's button assignments. Use the gesture reference next, then your game's play card. For a difficult pose, see [Make the controls fit your hand](#).

Your gesture reference

Practice one movement at a time in **Glove Academy**. Your selected profile decides what each gesture does in a game; the play cards below show the mapping. This reference also includes game-specific finger poses and combinations beyond the sixteen Academy lessons. The glove is an illustration: the Controller tracks your bare hand.

Resting position and movement

Gesture	See it	Try it
Show an open hand		Open your fingers comfortably and keep your whole palm visible.
Find your resting position		Hold a relaxed open hand at your saved centre and distance.
Move left		Slide your whole hand left.
Move right		Slide your whole hand right.
Move up		Raise your whole hand.
Move down		Lower your whole hand.

With an ordinary FCEUmm joystick profile, combine horizontal and vertical movement for diagonals. Your box is anchored to the hand center you saved during calibration; inside it-or exactly on its boundary-all positional directions stop immediately. Its effective size is never less than 1.5 times your saved hand size. Change its chosen size in Setup.

Wrist and depth movements

Gesture	See it	Try it
Roll wrist left		Tilt your hand left at the wrist.
Roll wrist right		Tilt your hand right at the wrist.
Push toward camera		Push deliberately, then return. Holding your hand nearby does not trigger a new Glove Zap.
Pull away from camera		Pull away deliberately, then return. Holding your hand far away does not trigger a new Pull Back.

Finger and hand poses

Gesture	See it	Try it
Curl index finger		Bend your index finger toward your palm.
Curl thumb		Fold your thumb across your palm.
Curl middle finger		Bend your middle finger toward your palm; used in Bad Street Brawler and Joust.
Curl ring finger		Bend your ring finger toward your palm; used in Defender II.
Close your hand		Curl your thumb and all four fingers into a comfortable fist.
Keep index straight		Keep your index finger extended for Gyruss fire.
Point index		Extend index and curl middle, ring, and pinky for the native Super Glove Ball Robo-Bullet.

Menu poses

Gesture	See it	Try it
V sign		Extend index and middle, curl ring and pinky, and hold steadily for about half a second for Start or pause.
Thumbs-up		Extend your thumb, close all four fingers, and hold until Select is recognized.
Menu guard		Curl thumb and ring only; keep index, middle, and pinky extended.

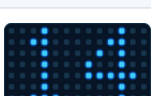
Start and Select send controller inputs; their effect depends on the game. The V sign and thumbs-up suppress A/B actions while forming, but some profiles can still produce movement. Keep your hand near its resting position. Use a physical controller if a game requires Select and a direction together.

Menu guard suppresses D-pad, A/B, Start, and Select, and cancels a pending Start press. It does **not** freeze native continuous hand positioning. For a dependable pause in all controller delivery while you reposition, use **Stop controller**.

Quick selector - Programs 1-14

These profiles reproduce the built-in Power Glove programs with camera gestures and ordinary NES controller output. Rapid fire defaults on only where an individual program description explicitly identifies a rapid or pulsed action. Among Programs 1-14, that means Program 7 starts with Rapid A on; the remaining numbered programs start with held A/B controls. Program 14 stops camera tracking and sends no VirtualGlove input so a conventional controller can be used without ending the authenticated game session.

Program	See it	Core controls	Games in Mattel's index
1		Hand position is the D-pad; thumb is A; index is B; curling the last three fingers performs one bounded turn-and-B action.	Blades of Steel; Blaster Master; Bubble Bobble; Castlevania; Castlevania II: Simon's Quest; Contra; Deadly Towers; Donkey Kong Classics; Double Dribble; Gauntlet; Gadius; Jackal; Kid Icarus; Kung-Fu Heroes; Metal Gear; Metroid; Mickey Mousecapade; Operation Wolf; Platoon; Racket Attack; Rampage; RoboWarrior; Rygar; Seicross; Star Force; Superman; Xenophobe; Zelda II: The Adventure of Link
2		Program 1 movement and buttons, plus live Centered / Return to centre feedback in place of the physical glove's beeper.	Centering-practice alternative; no indexed title
3		Push / pull sends Up / Down; side movement sends Left / Right; thumb / index sends A / B.	Ice Hockey; Top Gun. Gauntlet may also use this as an alternative top-view layout, although the index assigns it to Program 1.
4		Open or close all four fingers for Up / Down; index-versus-last-three poses steer the treads; thumb sends A; wrist poses send held B or Up plus Left / Right.	Iron Tank
5		Push / pull and bank / side movement fly the craft; thumb / index sends A / B.	Alpha Mission; Life Force; Xevious; 1943
6		Position and depth move; index sends A; thumb sends B; last three sends A+B; forward fist holds Up; clockwise twist turns rapidly twice.	Double Dragon
7		Open-hand movement dodges and ducks; forward fists punch high or low; clockwise wrist blocks; pull-back fist sends Select; thumb sends rapid A.	Mike Tyson's Punch-Out!!
8		Position and depth select bases or fielding direction; thumb, index, counter-clockwise wrist, and pull-back produce the documented offense / defense A and B actions.	Baseball; Bases Loaded; R.B.I. Baseball
9		Make a fist to ready the profile. Wrist rotation steers; forward fist is turbo; raising the hand sends Down; lowering sends B; fist sends A. No rapid fire.	Rad Racer

Program	See it	Core controls	Games in Mattel's index
10		Index / last-three curls steer Left / Right; thumb sends A; B is held automatically until the hand is lowered.	R.C. Pro-Am
11		Program 1 controls, but the last-three-finger pose turns rapidly in both directions while firing B.	Fast-turn alternative; no indexed title
12		Position moves Mario; thumb holds A for controllable jumps; index sends B; middle curl adds B for fast travel; last-three curl slows horizontal travel.	Super Mario Bros.
13		Thumb / index sends A / B. Camera D-pad output is neutral, leaving movement and menus to a conventional controller.	No indexed title
14		VirtualGlove camera and output are paused while the selected profile and game session remain visible.	Anticipation; temporary manual menu or password entry

The shipped exact-filename registry supplies [.nes](#), [.zip](#), and [.7z](#) entries. It retains Mattel's explicit rapid-fire-off entries for Alpha Mission, Blaster Master, Ice Hockey, Double Dribble, and Racket Attack. These entries are now redundant with the source-accurate profile defaults, but preserve the official game-specific instructions if a profile default changes later.

How rapid fire behaves

Original-hardware source note: Mattel's *Power Glove Instructions*, page 14, states, "Rapid Fire for both A and B turns on automatically when you turn on the glove." The same page immediately warns that not every glove program has rapid fire and directs players to the individual program descriptions. VirtualGlove therefore does not reproduce that blanket power-on state: it uses the documented behavior of the active profile instead.

Rapid A/B defaults come from the individual program descriptions rather than the manual's general power-on behavior. Program 7 defaults to Rapid A; Program B defaults to Rapid A for its pulsed flap; Program H defaults to both; and Bad Street Brawler defaults to Rapid B for its pulsed thumb action. Every other profile starts with A and B held. Documented compound, pulsed-direction, and turbo actions keep their own timing independently of these switches.

In VirtualGlove, **Rapid A** and **Rapid B** mean only that the corresponding NES button repeats while its gesture remains active. A profile's fast turns, pulsed steering, turbo movement, simultaneous-button combinations, and other compound actions are part of that profile and do not change when these two switches are changed.

Program 12 therefore holds A for as long as the thumb stays curled, allowing variable-height jumps in Super Mario Bros. If Rapid A is explicitly enabled on Dashboard, it repeats 250 ms A holds separated by a short rapid-fire gap. A registered game's settings travel in its signed launch request, and Dashboard's **Rapid fire** status shows the values actually applied to the running game.

After an upgrade: explicit Rapid A/B choices already saved for a game remain in force and override the corrected profile defaults. On Dashboard, choose **Use profile defaults** if you want that game to follow the source-accurate defaults again. This removes only its Rapid A/B override; it does not change the game's profile or any player settings.

Game	Program	Automatic exception	Why it matters while playing
Blaster Master	1	Rapid A off	A can be held continuously while thumb curl remains active.
Double Dribble	1	Rapid A and B off	Both finger buttons behave as held controls.
Racket Attack	1	Rapid A and B off	Both finger buttons behave as held controls.
Ice Hockey	3	Rapid B off	Retains Mattel's explicit held-B instruction.
Alpha Mission	5	Rapid A off	Retains Mattel's explicit held-A instruction.

Start, Select, Menu Guard, calibration, tracking-loss release, and the saved joystick dead zone remain shared across every camera-active numeric Program. Return to an open hand near the saved center between compound actions.

Program cards 1-14

Each card follows the same design as the game-specific cards: the exact Dashboard profile and matrix display, a mission, illustrated controls, practical play advice, and a first-round exercise. Programs without an official indexed title remain complete, selectable profiles and are especially useful for off-script experiments.

Program 1 - positional control

Profile	See it
program_1	

Your mission: Use the original all-purpose mapping to move in every direction, operate both action buttons, and learn its bounded turn-and-fire combination.

Do this	See it	Controller result
Move the whole hand		Ordinary D-pad, including diagonals
Curl the thumb		A
Curl the index finger		B
Curl middle, ring, and pinky together		Briefly turns opposite the last horizontal direction and sends B
Return to the saved center		Releases positional movement

Play smart: The turn-and-B combination lasts about 0.18 seconds and triggers once per fresh last-three-finger pose. Release those fingers before trying it again. A and B are held by default; a registered game can still opt into rapid behavior from Dashboard. The complete official Program 1 game list is in the numbered [Quick selector](#).

First round:

1. Move across all four sides of the saved center and return to neutral.
2. Try thumb A and index B separately.
3. Curl the last three fingers once, release them, and confirm the bounded turn-and-B action ends.

Program 2 - positional control with centering feedback

Profile	See it
program_2	

Your mission: Keep an exact sense of neutral while using Program 1's movement, buttons, and bounded turn-and-fire action in an unassigned game.

Do this	See it	Controller result
Move the whole hand		Ordinary D-pad, including diagonals
Curl the thumb		A
Curl the index finger		B
Curl middle, ring, and pinky together		Briefly turns opposite the last horizontal direction and sends B
Return to the saved center		Releases movement and shows Centered ; leaving the box shows Return to centre

Play smart: Program 2 has no indexed title and is the centering-practice alternative to Program 1. Dashboard's live feedback replaces the physical glove's beeper and stores no new centering measurement. Its turn-and-B action uses the same 0.18-second fresh-pose and release rules.

First round:

1. Rest inside the saved box and confirm Dashboard shows **Centered**.
2. Cross one side of the box, return until **Centered** reappears, then repeat on the other sides.
3. Try A, B, and one released turn-and-B action without losing the neutral reference.

Program 3 - depth and side movement

Profile	See it
program_3	

Your mission: Combine sideways hand travel with deliberate push/pull depth control for Ice Hockey, Top Gun, or a similar top-view game.

Do this	See it	Controller result
Move left or right		Left or Right
Push toward the camera		Up
Pull away from the camera		Down
Curl the thumb		A
Curl the index finger		B

Play smart: Use Program 3 for **Ice Hockey** and **Top Gun**. Ice Hockey automatically disables rapid B. Mattel also described Program 3 as an alternative top-view scheme for Gauntlet, but the official index and shipped registry assign Gauntlet to Program 1.

First round:

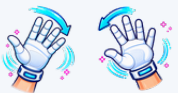
- 1. Move left and right while keeping a steady camera distance.
- 2. Push once for Up, return to neutral depth, then pull once for Down.
- 3. Add thumb A and index B only after the four directions release reliably.

Program 4 - Iron Tank tread control

Profile	See it
program_4	

Your mission: Drive Iron Tank with finger-tread poses, fire independently, and use the higher-priority wrist actions without leaving a direction held.

Program 4 does not use ordinary camera-position D-pad movement. Its finger and wrist poses drive the tank directly, in the following priority order.

Do this	See it	Controller result
Keep index, middle, ring, and pinky open		Up
Curl all four fingers		Down
Keep index open and curl the last three; or curl index and keep the last three open		Right tread turn; or Left tread turn, respectively
Roll wrist right or left		Up+Right; or Up+Left, respectively
Turn the wrist almost upside down		B
Curl the thumb		A alongside the active movement, except while pulling back
Pull away from the camera		Neutralizes the Program 4 action output

Play smart: Program 4 is assigned to **Iron Tank**. Release a wrist pose before changing to a finger-tread pose so the higher-priority wrist action clears first. Pulling back is the dependable way to neutralize its special action output.

First round:

1. Test open-four Up and closed-four Down, returning to a relaxed hand between them.
2. Try each asymmetric tread pose separately.
3. Add thumb fire, then practise each wrist action and pull back to neutralize it.

Program 5 - aircraft control

Profile	See it
program_5	

Your mission: Fly with horizontal movement, wrist bank, and depth while keeping both firing gestures available.

Do this	See it	Controller result
Move or bank left/right		Left or Right; a wrist bank can add the same direction
Push toward the camera		Up
Pull away from the camera		Down
Curl the thumb		A
Curl the index finger		B

Play smart: Use Program 5 for **Alpha Mission**, **Life Force**, **Xevious**, and **1943: The Battle of Midway**. Keep the wrist level when you want plain side movement; a bank can reinforce the same direction. Alpha Mission automatically disables rapid A.

First round:

- 1. Fly left and right with a level wrist.
- 2. Push and pull once each, returning to the saved distance after both actions.
- 3. Add thumb A, index B, and then a small wrist bank while moving.

Program 6 - Double Dragon combinations

Profile	See it
program_6	

Your mission: Chain Double Dragon's movement, single-button attacks, combined attack, forward-fist action, and bounded turning combination without carrying one pose into the next.

Do this	See it	Controller result
Move left/right; push/pull		Left/Right; push is Up and pull is Down
Curl the index finger		A
Curl the thumb		B
Curl middle, ring, and pinky together		A+B
Make a fist and push		Up only; A and B release
Roll wrist right		A bounded four-phase Left/Right/Left/Right turn over about 0.32 seconds

Play smart: Program 6 is assigned to **Double Dragon**. The wrist combination begins on a fresh right-roll hold; release the roll before repeating it. The pushed fist deliberately releases A and B while holding Up.

First round:

1. Walk in four directions, then test index A and thumb B separately.
2. Curl the last three fingers for A+B and fully reopen them.
3. Try one pushed fist and one right-roll combination, returning to neutral after each.

Program 7 - Punch-Out!! offense and defense

Profile	See it
program_7	

Your mission: Use open-hand defense and positioned fist attacks to complete a controlled round in Mike Tyson's Punch-Out!!.

Do this	See it	Controller result
With four fingers open, move left/right or lower the hand		Dodge Left/Right or duck Down
With four fingers open, curl the thumb		Rapid A
Make a fist, push, and hold it right of center		A punch; raising it above center adds Up for a high punch
Make a fist, push, and hold it left of center		B punch; raising it above center adds Up for a high punch
Roll wrist right		Down block
Make a fist and pull back		One Select pulse for the star-punch action

Play smart: Program 7 is assigned to **Mike Tyson's Punch-Out!!** and is the only numbered Program whose individual description explicitly calls for a rapid button default. Keep the fist clearly left or right of center so the intended punch is unambiguous. The pull-back Select is edge-triggered: return from the pull before requesting another star punch.

First round:

1. With an open hand, dodge left and right and lower the hand to duck.
2. Push a fist on each side for low punches, then raise it for high punches.
3. Practise one wrist block and one pull-back star-punch pulse.

Program 8 - baseball offense and defense

Profile	See it
program_8	

Your mission: Learn when the baseball mapping changes the thumb from B to direction-plus-A, then use the same profile for batting, running, and fielding.

Do this	See it	Controller result
Move left/right; push/pull		Left/Right; push is Up and pull is Down
Curl the thumb while stationary		B
Curl the thumb while moving		A alongside the active direction
Curl the index finger		B
Roll wrist left		B
Pull away from the camera		Down+B

Play smart: Use Program 8 for **Baseball**, **Bases Loaded**, and **R.B.I. Baseball**. The same thumb pose intentionally changes button meaning depending on whether a direction is active. Establish movement first, then add the thumb when you need direction-plus-A.

First round:

1. Move in each direction with an open thumb and return to neutral.
2. Curl the thumb while stationary to confirm B.
3. Move and curl the thumb for direction-plus-A, then test index B and pull-back Down+B.

Program 9 - Rad Racer

Profile	See it
program_9	

Your mission: Arm the Rad Racer profile, steer by wrist angle, and combine its held throttle, turbo, brake, and unusual raise-to-Down mapping.

Program 9 begins unready. Make one fist to arm it; Dashboard then shows the ready state for the remainder of that profile session. It has no rapid fire.

Do this	See it	Controller result after arming
Roll wrist left/right		Steer Left/Right
Keep a fist closed		A
Push the closed fist		Up+A turbo combination
Raise the hand		Down
Lower the hand		B

Play smart: Program 9 is assigned to **Rad Racer**. Its unusual raise-to-Down mapping is intentional and follows the documented program rather than the general movement convention. All inputs are held rather than pulsed.

First round:

1. Make one fist and confirm the profile becomes ready.
2. Hold the fist for A and make small wrist rolls to steer.
3. Push the fist for turbo, then separately raise for Down and lower for B.

Program 10 - R.C. Pro-Am

Profile	See it
program_10	

Your mission: Drive R.C. Pro-Am using finger-tread steering while hand height controls the automatically held B action.

Do this	See it	Controller result
Curl only the index finger		Left
Keep index open; curl middle, ring, and pinky		Right
Make both steering poses at once		Steering releases rather than pressing both directions
Curl the thumb		A
Keep the hand above the lower movement region		B is held automatically
Lower the hand outside the center box		Releases B

Play smart: Program 10 is assigned to **R.C. Pro-Am**. Hand position does not steer; use the two mutually exclusive finger poses. Curling both steering patterns releases direction instead of sending an impossible Left+Right pair.

First round:

1. Keep the hand above the lower region and confirm B is held.
2. Alternate the index-only and last-three steering poses, reopening between them.
3. Lower the hand to release B, then add thumb A when the steering feels reliable.

Program 11 - rapid turn alternative

Profile	See it
program_11	

Your mission: Apply Program 1's familiar movement and buttons to an unassigned game, then decide whether its sustained alternating turn-and-B action fits the game's mechanics.

Do this	See it	Controller result
Move the whole hand		Ordinary D-pad, including diagonals
Curl the thumb		A
Curl the index finger		B
Curl middle, ring, and pinky together		Overrides positional movement, alternates Left and Right, and sends B while held

Play smart: Program 11 has no indexed title. It is a Program 1-style alternative for games where a sustained rapid-turn action is useful. Unlike Program 1's short turn, the alternating turn continues until the last three fingers are released, so test it briefly before relying on it.

First round:

1. Confirm ordinary hand-position movement and thumb/index buttons first.
2. Curl the last three fingers for a short fast-turn burst.
3. Reopen fully and verify movement and B release before trying a longer burst.

Program 12 - Super Mario Bros.

Profile	See it
program_12	

Your mission: Move Mario with hand position, combine action buttons for running and jumping, and deliberately switch between fast and slow travel.

Do this	See it	Controller result
Move the whole hand		Ordinary D-pad
Curl the thumb		Hold A for as long as the thumb remains curled
Curl the index finger		B
Curl the middle finger without curling all three last fingers		B for fast travel
Curl middle, ring, and pinky together while moving sideways		Pulses only the active horizontal direction for slower travel; vertical movement is unchanged

Play smart: Program 12 is assigned to **Super Mario Bros.** Rapid A is off by default, so jump height follows how long the thumb remains curled. If you manually enable Rapid A, a held thumb repeats 250 ms A presses separated by a short gap. Use middle curl for fast travel and the last-three pose only while you want slower horizontal movement. Release the last-three pose to return immediately to ordinary travel.

First round:

1. Walk in both directions, vary the length of the thumb curl to compare jump heights, and try index B separately.
2. Curl the middle finger while moving to practise fast travel.
3. Use the last-three pose during a safe horizontal stretch, then reopen to restore full movement.

Program 13 - finger buttons with physical movement

Profile	See it
program_13	

Your mission: Split Player 1 cleanly between gesture action buttons and the merged physical controller's movement and menus.

Do this	See it	Controller result
Move the hand anywhere		No camera D-pad output
Curl the thumb		A
Curl the index finger		B

Play smart: Program 13 has no indexed title. Use it when VirtualGlove should provide the two action buttons while the merged physical Player 1 controller supplies movement or menu combinations. Shared gesture Start, Select, and Menu Guard remain available, but camera movement never sends a D-pad direction.

First round:

- 1. Move the physical controller in all four directions and confirm hand movement stays neutral.
- 2. Keep the physical controller centered and try thumb A and index B separately.
- 3. Combine one physical direction with one gesture button, then release both.

Program 14 - gestures off for the active game

Profile	See it
program_14	

Your mission: Pause every VirtualGlove gesture safely while keeping the authenticated game session and physical Player 1 controller available.

Do this	See it	Controller result
Select Program 14		Camera tracking stops and every VirtualGlove control releases
Use the physical Player 1 controller		Physical D-pad and buttons remain available
Select any camera-active profile		Recognition starts again using saved calibration

Play smart: Program 14 is assigned to **Anticipation** and can be selected temporarily for manual menu or password entry. It preserves the visible profile and authenticated game session but neutralizes VirtualGlove D-pad, A, B, Start, and Select. No calibration or player data is erased.

First round:

1. Select Program 14 and confirm the camera closes and Dashboard controls are neutral.
2. Complete one menu action with the physical Player 1 controller.
3. Select a camera-active profile and confirm recognition returns with the saved player calibration.

Programs A-I

The original Power Glove could load nine reusable mappings from Bad Street Brawler and retain one while the player changed cartridges. VirtualGlove keeps all nine available at once: choose one on Dashboard or let RetroPie select one from the registered ROM filename. These mappings produce ordinary NES controller inputs, so they can be tried with games beyond the tested play cards.

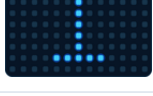
Where the programs came from

Bad Street Brawler contained nine configuration programs labelled A through I. The player loaded one into the Power Glove, switched off the NES, swapped to a different cartridge within roughly 30 seconds, and kept using the downloaded mapping.

These programs were not replacement firmware. They were small configurations for the glove's resident gesture interpreter. Each program mapped hand position, depth, wrist angle, and finger bends to ordinary NES controller inputs. The next game therefore did not need special Power Glove support.

VirtualGlove keeps all nine profiles ready at once. Select one on Dashboard or let RetroPie choose it when a game launches. You do not need to open Bad Street Brawler first.

Quick selector - Programs A-I

Program	See it	Best fit	Main controls
A		Pinball	Two finger flippers, wrist tilt, combined-flipper mode
B		Joust	Pulsed steering by position; curl a finger for a pulsed flap
C		Gyruss	Rotate by wrist angle; fire and bomb gestures
D		Challenge mode	Reversed directions with thumb/index buttons
E		Defender II	Ship movement, fire, smart bomb, evasive movement
F		Sesame Street 1-2-3	Open-hand Yes and closed-hand No
G		Gun Smoke	Walk by position; combine index curl and a forward push to fire
H		Training / general play	Familiar controls with pulsed buttons
I		Knight Rider / driving	Wrist steering, throttle, brake, and turbo

Program cards A-I

A - Pinball rig

Profile	See it
program_a	

Gesture	See it	Controller result
Curl the index finger		Right flipper / A
Curl the thumb		Left flipper / Up
Roll the wrist left or right		Tilt / B
Pull the hand away from the camera		Toggle combined-flipper behaviour

Use this profile for pinball tables and games that benefit from two independent finger actions.

B - Joust rig

Profile	See it
program_b	

Gesture	See it	Controller result
Move the hand left or right		Pulsed steering left or right
Curl the thumb		B button; see the Joust play card for its in-game use
Curl the index or middle finger		Pulsed flap input

Use this profile for Joust and any game where rhythmic, repeated presses matter.

C - Gyruss rig

Profile	See it
program_c	

Gesture	See it	Controller result
Roll the wrist left or right		Rotate counter-clockwise or clockwise
Keep the index finger straight		Continuous fire
Pull the hand away from the camera		Launch a bomb

Use this profile for circular shooters and games with rotation plus rapid fire.

D - Mirror-world rig

Profile	See it
program_d	

Gesture	See it	Controller result
Move the hand left or right		Reversed right or left direction
Raise or lower the hand		Reversed down or up direction
Curl the thumb		First action button
Curl the index finger		Second action button

Use this profile for deliberate chaos, party challenges, or accessibility experiments that need inverted direction mappings.

E - Defender rig

Profile	See it
program_e	

Gesture	See it	Controller result
Move the whole hand		Move the ship
Curl the thumb		Fire
Roll the wrist left or right		Smart bomb
Curl the ring finger		Rapid horizontal movement

Use this profile for Defender II and multi-action shooters.

F - Yes / No rig

Profile	See it
program_f	

Gesture	See it	Controller result
Close every finger into a fist		No
Move an open hand in any direction		Yes

Use this profile for Sesame Street 1-2-3 and simple choice-driven games.

G - Gun Smoke rig

Profile	See it
program_g	

Gesture	See it	Controller result
Move the whole hand		Walk using horizontal and vertical hand movement
Curl the index finger		Fire
Roll the wrist		Add left or right movement
Curl the thumb and ring finger		Suppress all ordinary controller output

A forward push sends B. Combine it with index curl (A) to send A+B.

Use this profile for Gun Smoke and shooters with movement plus directional fire.

H - Training rig

Profile	See it
program_h	

Gesture	See it	Controller result
Move the whole hand		Conventional directions
Curl the thumb		Pulsed A
Curl the index finger		Pulsed B
Return the hand to center		Release directional input

Use this profile for learning the system or giving an unmapped game a sensible general-purpose starting point.

I - Driving rig

Profile	See it
<code>program_i</code>	

Gesture	See it	Controller result
Roll the wrist left or right		Steer
Curl the index finger		Throttle
Lower the hand		Brake
Push the hand toward the camera		Turbo
Curl the thumb		Auxiliary action

Use this profile for Knight Rider and driving games that need steering, speed, and one extra action.

Bad Street Brawler

Profile	See it
bad_street_brawler	

Your mission: Guide Duke Davis through each stage, discover that stage's three fighting moves at the practice bag, and clear the street before time or vitality runs out.

Do this	See it	Duke does this
Move hand left / right		Walk left / right
Raise / lower hand		Jump / crouch
Curl thumb		Pulsed B move
Curl middle finger		A+B force move
Roll wrist left / right		A plus that direction
Push toward camera		Glove Zap

Play smart: The available force moves change with each stage. Test the thumb curl, middle-finger curl, and wrist rolls on the punching bag before leaving practice. Push toward the camera to trigger Glove Zap, then return to your starting distance before another attempt. The game controls its once-per-round availability. If Zap does not work, check the [game-specific setup](#).

First round:

1. At the practice bag, try a thumb curl, a middle-finger curl, and a wrist roll separately.
2. Notice which move each gesture produces in this stage.
3. Enter the street and use one familiar move before adding combinations.

Super Glove Ball

Profile	See it
super_glove_ball	

Your mission: Control the Robo-Glove, keep the energy ball in play, break a complete wall of tiles, and follow the revealed arrows through the maze.

Nestopia (VirtualGlove) means the custom [lr-nestopia-powerglove](#) core. Use the controls below for the emulator that actually starts; FCEUmm is the fallback. The RetroPie launch hook reports that running core automatically. Only Super Glove Ball running in [lr-nestopia-powerglove](#) uses native input. Starting the same ROM in FCEUmm-or in any other or unknown core-keeps joystick output active for the whole session. The V sign sends Start in both modes.

Do this	See it	Controller result
Move whole hand		FCEUmm: eight-direction digital steering. Nestopia (VirtualGlove): continuous hand positioning.
Curl index finger		FCEUmm: A, move the glove into the room.
Curl thumb		FCEUmm: B, punch, grab, or launch a new ball.
Open hand		Nestopia (VirtualGlove): release or throw a held ball.
Close hand		Nestopia (VirtualGlove): grab or catch the ball.
Point index; curl middle, ring, and pinky		Nestopia (VirtualGlove): fire a Robo-Bullet.
Close hand and push forward		Nestopia (VirtualGlove): Power Punch.

Play smart: Pick one wall and finish it, then follow the revealed arrow. With FCEUmm, use Select to take the exit. Native gameplay has been completed without Select; although the core can transmit its native button bit, no required in-game action has been identified for it. **Latest coordinate** is the native movement default. It follows each newest valid palm point directly during continuous tracking and waits one fresh result only for a contradictory or unusually distant non-forward reacquisition. Continuous native movement is playable and has been substantially tightened, although synchronized physical latency measurement is still pending. The [native compatibility record](#) contains the test evidence and details about additional packet fields.

First round:

1. Move the Robo-Glove across the room with small hand movements.
2. With **Nestopia (VirtualGlove)**, practise closing to grab and opening to release. Try index-point fire and a fist-plus-forward Power Punch separately.
3. With **FCEUmm**, try the index-curl A action and thumb-curl B action separately.

Joust

Profile	See it
program_b	

Your mission: Ride the ostrich, strike enemy riders from above, collect their eggs before they hatch, and stay clear of the lava.

Do this	See it	Result
Move hand left / right		Pulsed steering left / right
Curl index or middle finger		Pulsed A: steady flap
Curl thumb		B: faster flap
Hold thumbs-up		Select game mode

Play smart: Height wins jousts. Use the faster thumb flap to climb, then the pulsed finger flap to hold position. Sweep up eggs quickly; every ignored egg is an enemy preparing a return engagement.

First round:

1. Curl your index finger to practise a steady flap.
2. Move left and right while keeping your height.
3. Approach one rider from above, then collect the egg.

Gyruss

Profile	See it
program_c	

Your mission: Circle the tunnel, destroy incoming formations, survive the warp zones, and fight from planet to planet toward the Sun.

Do this	See it	Result
Roll wrist left / right		Orbit counter-clockwise / clockwise
Keep index finger straight		Continuous A fire
Pull hand away from camera		B bomb
Hold thumbs-up		Select control mode

Before launching: Choose **Attack Control B** at the title screen. This profile expects left/right rotation rather than eight-direction movement.

First round:

1. Select Attack Control B at the title screen.
2. Keep your index straight and practise small wrist rolls in both directions.
3. Clear one formation before trying the pull-back bomb.

Defender II

Profile	See it
program_e	

Your mission: Patrol the planet, destroy alien raiders, and rescue the humanoids before abductors carry them away and turn them into mutants.

Do this	See it	Result
Move whole hand		Fly up, down, left, or right
Curl thumb		A: fire
Roll wrist either way		B: smart bomb
Curl ring finger		Rapid left/right evasive thrash

Play smart: Watch the scanner as much as the ship. Intercept abductors early; if one lifts a humanoid, shoot the alien and catch the falling person. Wrist rolls trigger the smart-bomb action, so make them deliberate.

First round:

1. Fly a short circuit with your wrist level.
2. Curl your thumb to fire while moving.
3. Track one abductor on the scanner; save deliberate wrist rolls for smart bombs.

Sesame Street 1-2-3

The same game may appear in your library as **Sesame Street 123**.

Profile	See it
program_f	

Your mission: Play the counting activities by giving the game a simple, physical Yes or No answer.

Do this	See it	Result
Move an open hand in any direction		A: Yes
Close every finger into a fist		B: No

Play smart: Directional output is intentionally disabled in this profile. Make the open-hand answer broad and obvious; make the fist complete. Return to a relaxed open hand between questions so one answer does not run into the next.

First round:

1. Count the objects before making a gesture.
2. Move an open hand from the resting position for Yes, or close all fingers for No.
3. Return to a relaxed hand at the centre before the next question.

Gun Smoke

Profile	See it
program_g	

Your mission: Walk the scrolling frontier, defeat the bandits, find or buy each wanted poster, and collect the bounty by beating the stage boss.

Do this	See it	Result
Move whole hand		Walk in that direction
Curl index finger		A: shoot diagonally right
Push toward camera		B: shoot diagonally left
Curl index while pushing		A+B: shoot straight ahead
Curl thumb and ring finger		Suppress D-pad, A/B, Start, and Select

Play smart: A stage keeps looping until you obtain its wanted poster. Keep your palm level while walking; wrist roll can add a left/right movement command. Use index-plus-push when you need the straight-ahead shot.

First round:

1. Try an index curl for the right shot and a forward push for the left shot.
2. Combine them to fire straight ahead.
3. Walk while firing, then look for the wanted poster.

Knight Rider

Profile	See it
program_i	

Your mission: Drive KITT from city to city, avoid roadside hazards, destroy the criminals ahead, and reach each destination before the timer expires.

Do this	See it	Result
Roll wrist left / right		Steer left / right
Curl index finger		Accelerate
Lower hand		Brake
Push toward camera		Accelerate plus turbo boost
Curl thumb		Fire weapons

Play smart: Keep the wrist near center on straight roads; large steering rolls are for real turns. Keep your index finger curled for normal speed and reserve the forward push for a clean burst when the road opens.

First round:

1. Curl your index finger to accelerate and make small wrist rolls to steer.
2. Lower your hand to practise braking.
3. Use a forward push for turbo only when the road ahead is clear.

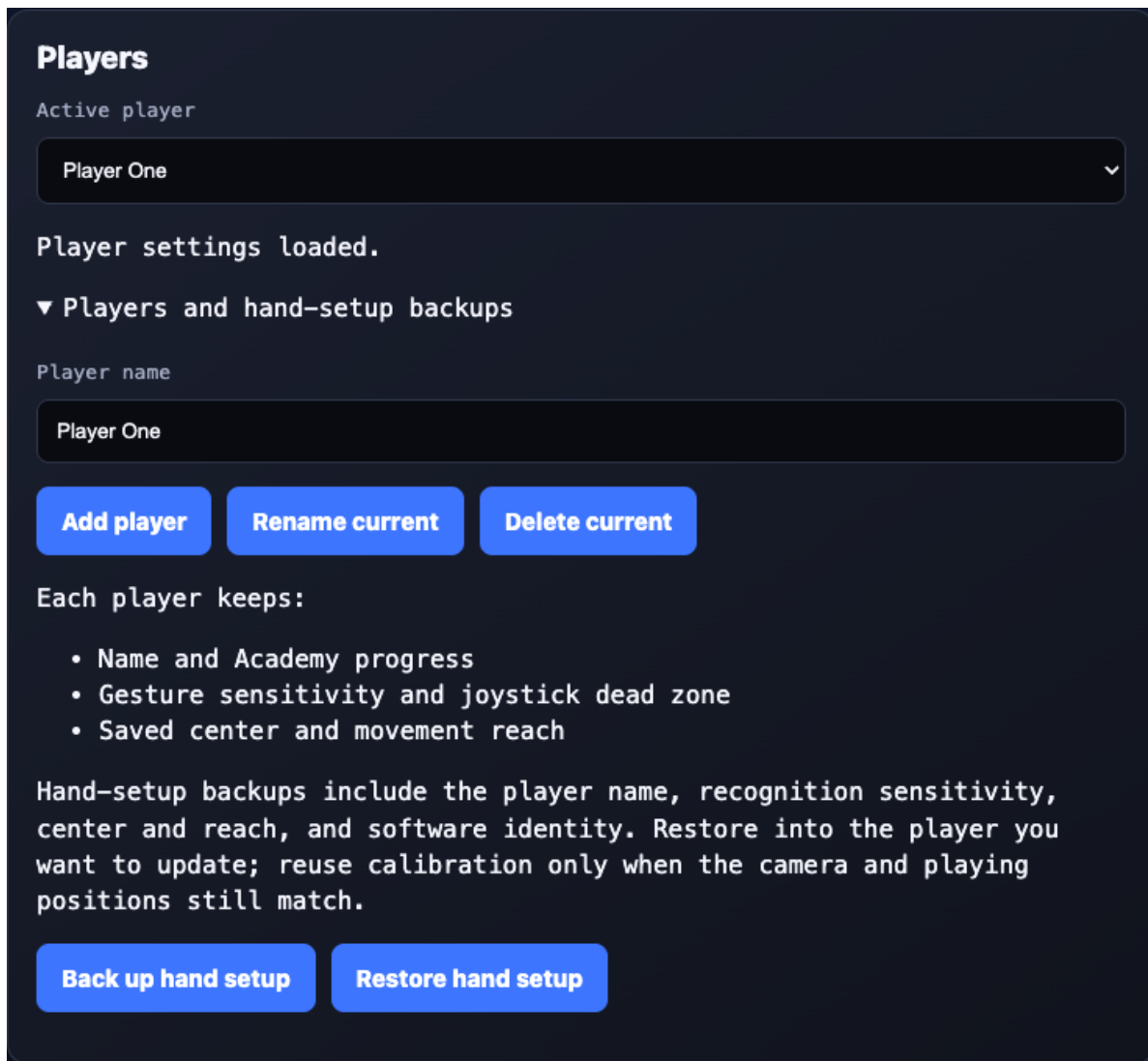
Make the controls fit your hand

Choose a player and save your setup

Choose a player in **Active player** before practising or tuning. Each player keeps separate sensitivity, lesson progress, and a saved centre. Selecting a player loads all three automatically and pauses controller output. Use **Center hand** for players without a saved centre or after moving the camera or changing playing position.

Use **Setup -> Players -> Players and hand-setup backups -> Back up hand setup** to save the player's name, center-box size, personal and complete gesture sensitivity, software identity, and calibration. During restore, choose whether to keep the complete saved sensitivity, including the defaults used when the backup was made, or just personal adjustments. Reuse calibration only with the same camera and playing position; otherwise set a fresh centre. Backups do not include credentials or Academy progress. New exports use the [virtualglove-hand-setup](#) format at version 4. Older backup formats are rejected without changing the selected player.

Choose each player in turn and select **Back up hand setup** to download a separate file named for that player, such as [alex-virtualglove-hand-setup.json](#). Your browser saves it on the computer, phone, or tablet you are using, usually in **Downloads** or the folder you choose. To restore, select the player you want to update, choose **Restore hand setup**, and pick that player's saved file from your device. Review it before confirming; restore updates the selected player, rather than adding a new one.

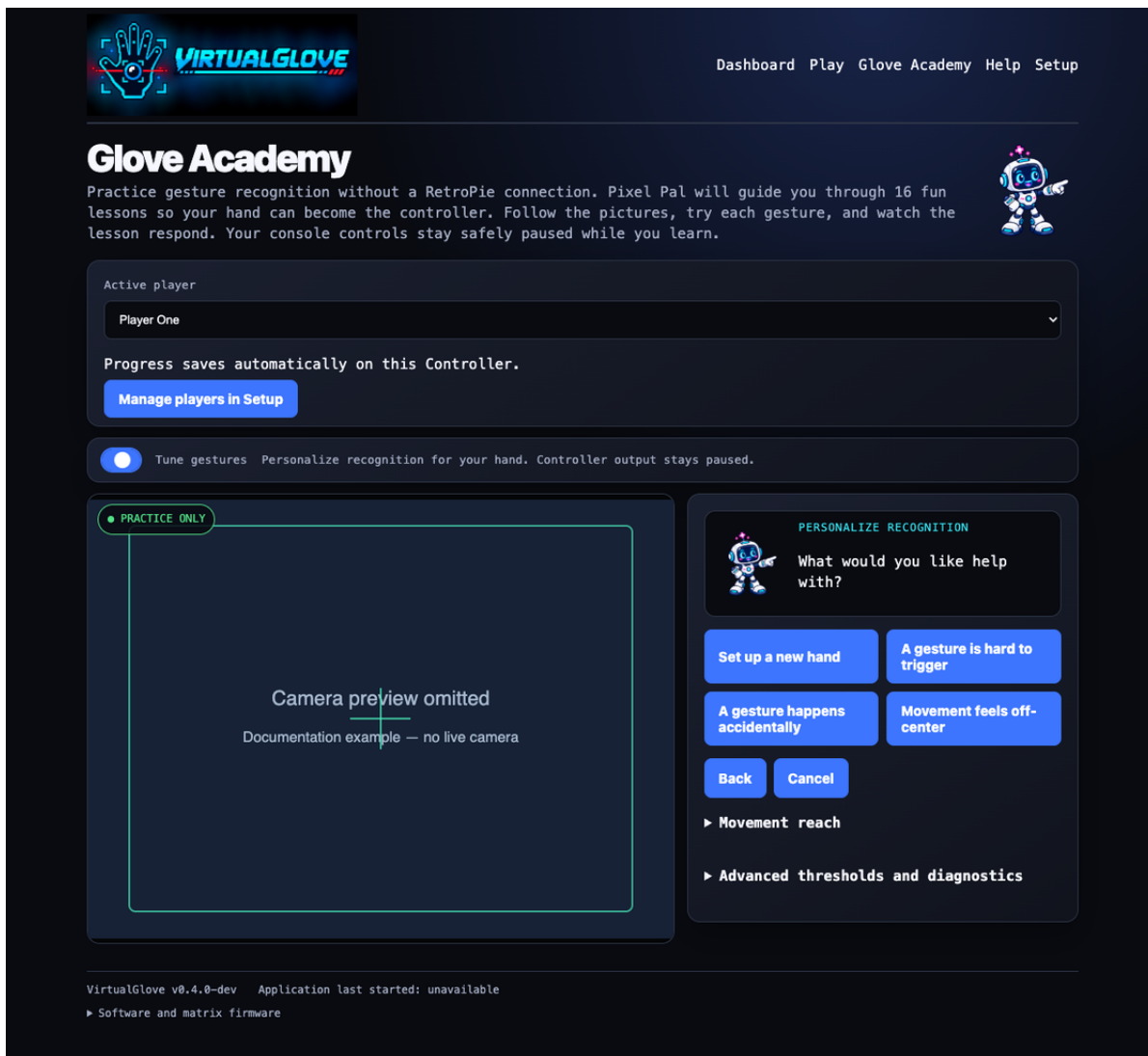


Player selection and complete hand-setup backups

Tune a difficult gesture

Tuning is optional. In **Glove Academy**, switch on **Tune gestures**. Pixel Pal helps with a new hand setup, a difficult or accidental gesture, or movement that feels off-centre. This adjusts recognition thresholds, not a personal camera model. It does not change the game's button assignments.

1. Choose what you want to improve and follow the framing advice.
2. When your whole hand is tracked clearly, select **I'm ready**.
3. Follow the countdown and prompts. Finger poses use open hand, performed pose, then open hand again. Movement uses your starting position, the motion, then a return. Forward and backward movements are repeated three times.
4. Test the result with two clean activations and releases, then hold neutral for three seconds. Save when the check passes, or retry the indicated pose.



Tune mode with Pixel Pal guiding the personalization choices

Cabinet output stays paused during practice and tuning. The live camera area is excluded from these screenshots for privacy. Tune one difficult gesture without repeating the whole hand setup. Pixel Pal may suggest better framing or lighting; it does not change camera exposure automatically.

Use the separate **Movement reach** section when native Super Glove Ball needs more or less physical travel. Left, right, up, and down are normalized distances from the saved center; smaller values reach the corresponding screen edge sooner. The summary shows the tracking area's width, height, and aspect ratio. Saving changes only those four reach spans. **Restore full camera field** returns to the camera-boundary mapping without changing the center or gesture sensitivity.

Setup's advanced camera settings offer Automatic, 30 fps, and 60 fps. Automatic prefers the tested 30-fps path and falls back safely to a camera-supported rate; the active rate is shown while tracking runs. Leave it on Automatic unless a specific camera comparison is needed.

Exposure is also Automatic by default. If a compatible Linux UVC camera looks too dark or loses the hand during fast movement, Setup offers exposure controls. Direct V4L2 remains an engineering comparison with Manual exposure and gain. The page shows whether manual control was supported and which values were actually applied. The project's Kiyo Pro tested well at exposure 78 and gain 96, but lighting and camera models differ. Manual mode returns the camera to automatic exposure when tracking closes.

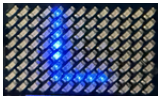
Leave **Advanced thresholds and diagnostics** closed unless you need numerical controls or a diagnostic run. The [Configuration Reference](#) explains the threshold checks and targeted restore options.

Check the result in a game

Return to Dashboard, select **Start controller**, and try the same gesture in a short game session. Check that it activates reliably and releases when you return to rest. If another gesture activates accidentally, stop delivery and revisit that pose in Glove Academy. Saved sensitivity is shared across your selected player's games; each game profile still decides the resulting actions.

Understand the matrix display

The matrix shows the active game or practice mode. Each play card includes its game display; the images below identify Academy and tuning.

Mode	See it	Controller output
Glove Academy lessons		Paused while you practise
Tune gestures		Paused while you record or preview thresholds

For the idle display, choose **Setup -> Matrix attract mode: On, Dim, or Off** with faint connection pixels. This affects only the idle animation, not game displays, Academy, tuning, or gesture recognition. The setting saves without restarting tracking. See the [Matrix display guide](#) for connection pixels, startup animations, and the full display reference.

How VirtualGlove selects a program

Choose the current profile on Dashboard; choose the saved startup profile on Setup. Automatic selection matches the complete ROM filename, including its extension but excluding its folder path, against [/etc/virtualglove/games.json](#) on RetroPie. Matching ignores letter case.

The launch hook sends an authenticated profile request. The VirtualGlove Controller releases held controls, changes the mapping, reuses the saved calibration, and acknowledges the new profile on its blue matrix. If no valid calibration is saved, it collects an initial reference while you hold your open hand still in a comfortable resting position. It uses 24 geometrically valid observations; repeating the same center, distance, and wrist pose produces a similar rather than bit-for-bit identical reference.

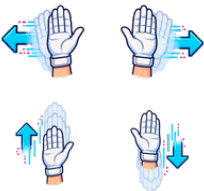
```
JSON
{
  "games": {
    "Joust (USA).zip": "program_b",
    "Gyruss (USA).nes": "program_c",
    "Gun.Smoke (USA).7z": "program_g"
  }
}
```

Merge entries into the existing [games](#) object; do not replace other registered games. The matrix displays **1** through **14** or **A** through **I** for those profiles. When an unknown game starts, the launch hook turns gesture control off so the previous game's mapping does not remain active.

Take VirtualGlove off-script

You can use the included profiles with games beyond Mattel's indexed titles and the dedicated play cards in this guide. Programs 1-13 and A-I send ordinary NES controller inputs, so try matching their gestures to games with similar controls. Programs 2, 11, 13, A, D, and H have no default ROM assignment and are useful starting points for these experiments. Program 14 is the deliberate camera-and-output-off choice rather than a general gesture mapping.

Start with 2, 11, 13, A, D, and H

Program	See it	Try it with	Know before playing
2 - Centering coach		Maze, platform, and precision-movement games where a clear neutral position helps	It uses Program 1's D-pad and A/B gestures, while Dashboard adds live Centered / Return to centre feedback.
11 - Fast turn		Arena action and shooters that benefit from a repeated left/right turn while firing	Hand position and thumb/index curls follow Program 1. Curling the last three fingers alternates Left/Right and fires B until released.
13 - Finger buttons		Games where you want gesture A/B while keeping movement and menus on the physical Player 1 controller	Thumb and index curls provide A and B. Camera movement sends no D-pad, so the merged physical controller remains in charge of direction.
A - Pinball		Pinball and games driven by two independent actions	Index curl is A, thumb curl is Up, wrist tilt is B, and pulling back toggles combined flippers. Ordinary directional movement is disabled.
D - Mirror world		A game you already know well, a party challenge, or an inverted-direction accessibility experiment	Every direction is reversed. Thumb and index provide A and B. Expect your muscle memory to complain loudly.
H - General play		Two-button platform, maze, puzzle, and action games	Hand movement supplies the D-pad. Thumb pulses A and index pulses B, so games that require a long held button may be a poor fit.

Try Program H for general play, Program 2 when neutral-position feedback would help, or Program 13 when you want the physical controller to handle movement. Program 11 supplies a sustained fast turn, Program A covers pinball controls, and Program D turns a familiar game into a new coordination challenge without changing the ROM or emulator.

Try a combination

1. Launch the NES or Famicom game normally. An unregistered game safely turns gesture output off instead of inheriting the previous game's controls.
2. Open the VirtualGlove Controller **Dashboard** and choose **2: Centering coach**, **11: Fast turn**, **13: Finger buttons**, **A: Pinball**, **D: Challenge**, **H: General**, or another Program 1-14 or A-I profile from **Active profile**.
3. Use **Center hand** if your resting hand position produces unwanted movement or your physical setup has changed. Hold a relaxed open hand still at your intended centre and distance until calibration finishes, then select **Start controller** and return to the game.
4. Test movement, both action gestures, Start, and Select before committing to a long session. Stop the controller immediately if a gesture remains active.

The selection is temporary. Starting or ending a game sends a new command that changes the profile or turns gestures off.

Keep a discovery

Open **Setup -> Games** and map the game's exact ROM filename to the profile that worked. Include its **.nes**, **.zip**, or **.7z** extension, validate, and save. Restart the game to use the new mapping. Automatic selection is currently limited to NES and Famicom; other systems turn gesture control off until their mappings and launch behaviour are validated.

Games

Edit the game mappings installed on your RetroPie. Saving affects the next game launch.

Use the exact ROM filename, including its extension: **Joust (USA).7z** and **Joust (USA).nes** need separate entries. Matching ignores letter case. Only NES and Famicom launches use these mappings.

► Available profile identifiers

Game mappings JSON

```
{
  "games": {
    "Bad Street Brawler (USA).nes": "bad_street_brawler",
    "Bad Street Brawler (USA).zip": "bad_street_brawler",
    "Bad Street Brawler (USA).7z": "bad_street_brawler",
    "Super Glove Ball (USA).nes": "super_glove_ball",
    "Super Glove Ball (USA).zip": "super_glove_ball",
    "Super Glove Ball (USA).7z": "super_glove_ball",
    "Joust (USA).nes": "program_b",
    "Gyruss (USA).nes": "program_c",
    "Defender II (USA).nes": "program_e",
    "Sesame Street 1-2-3 (USA).nes": "program_f",
    "Gun.Smoke (USA).nes": "program_g",
    "Knight Rider (USA).nes": "program_i",
    "Joust (USA).zip": "program_b",
    "Joust (USA).7z": "program_b",
    "Gyruss (USA).zip": "program_c",
    "Gyruss (USA).7z": "program_c",
    "Defender II (USA).zip": "program_e",
    "Defender II (USA).7z": "program_e",
    "Sesame Street 1-2-3 (USA).zip": "program_f",
    "Sesame Street 1-2-3 (USA).7z": "program_f",
    "Gun.Smoke (USA).zip": "program_g",
    "Gun.Smoke (USA).7z": "program_g"
  }
}
```

Validate

Format

Save

Reload

Download backup

Restore previous save

Installed mappings loaded.

Games mappings in the lower section of Setup

Do not stop at A, D, and H. Try B when rhythmic button pulses suit the action, C when wrist rotation can replace horizontal movement, F for simple Yes/No choices, or I when steering and throttle are the heart of the game. Match the profile to the game's mechanics, not to the title printed on the cartridge.

GLOVE LAB RULE A strange pairing that is controllable, repeatable, and fun is a successful experiment. Record the exact ROM filename and profile so somebody else can reproduce it.

For registry validation and manual profile commands, see the [Configuration Reference](#).

Sources, artwork, and fair play

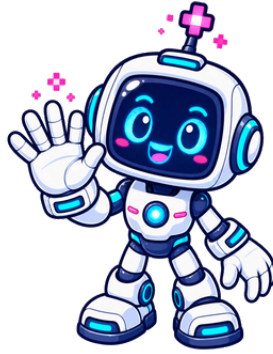
The profile descriptions are checked against the project's implemented gesture engine and tests. Game objectives and original control intent were summarized from the following historical instruction sources:

- Mattel, *Power Glove Instruction Manual* (complete manual; Program 4 and the official game index), supplied for this release's documentation review
- Mattel, *Power Glove Program Guide* (1988 US; program illustrations and the 1943 workflow), supplied for this release's documentation review
- [Mattel Power Glove instructions and Programs A-I](#)
- [Bad Street Brawler NES instruction transcription](#)
- [Super Glove Ball NES instruction manual](#)
- [Joust NES instruction transcription](#)
- [Gyruss NES instruction transcription](#)
- [Defender II NES instruction transcription](#)
- [Gun Smoke NES gameplay reference](#)
- [Knight Rider NES instruction manual](#)

The gesture drawings are original VirtualGlove project illustrations made for this guide. They deliberately avoid game screenshots, box art, characters, and publisher logos.

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Pixel Pal's Extra-Digit Hunt answer



Pixel Pal reveals the Extra-Digit Hunt answer

Pixel Pal's answer: 28 six-digit hands.

One appears in the opening Rock, Paper, Scissors table. Ten appear in the numbered Program cards: Programs 1, 2, 4, 6, 7, 9, 10, and 11, with Program 12 appearing twice. Two more appear in the Program 11 and Program 13 off-script cards. The remaining fifteen appear in the shared gesture reference, dedicated game cards, Programs B, E, and F, and the Program H off-script card. Every appearance counts, even when the same artwork returns.